MAT 120 Lab – Week 1

Introduction to Function Visualization with Mathematica

Department of Mathematics and Physical Sciences

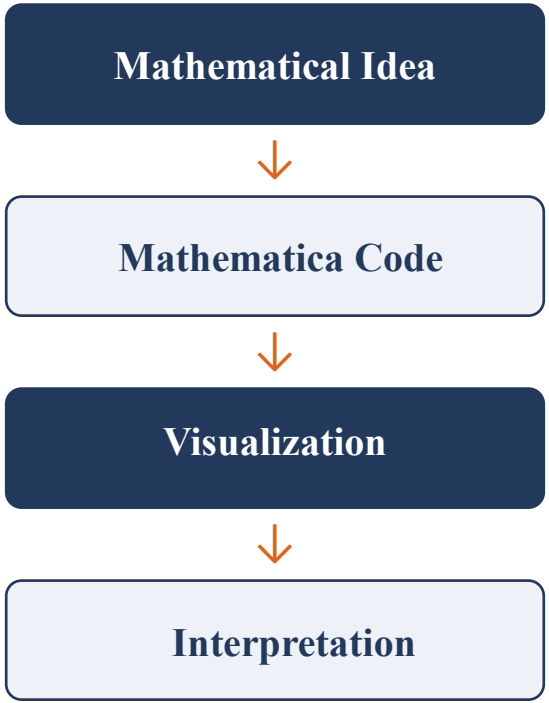
BRAC University

From a formula → a function → a graph → mathematical meaning.

What this Lab is really about

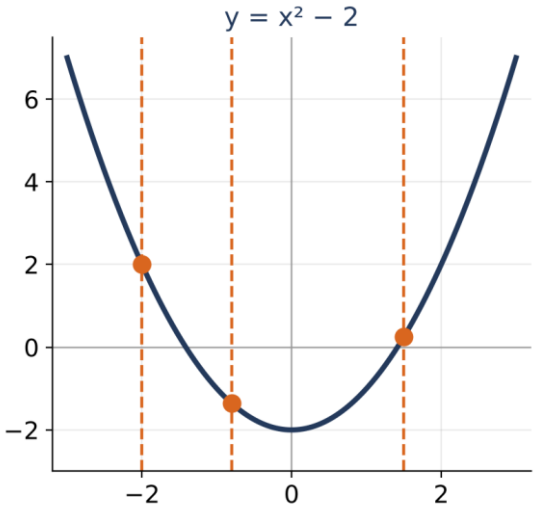
By the end of today, you should be able to

- Read input, output, domain and range;
- Recognize increasing/decreasing and positive/negative behavior;
- Compare polynomial, trigonometric, rational and piecewise functions;
- Use graphs to estimate intercepts, turning points and discontinuities;
- Distinguish exact answers from numerical approximations.

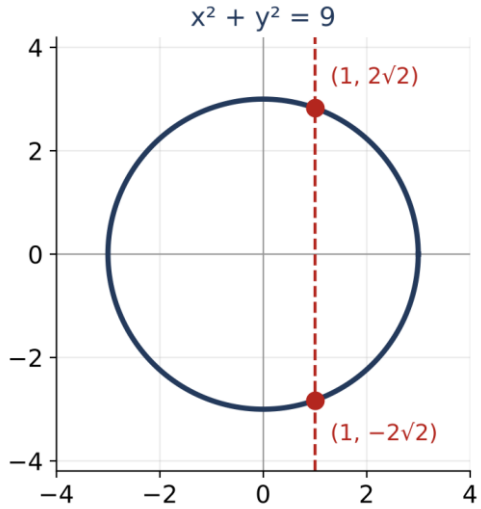


FUNCTION

✓ **Passes: this is a function**



✗ **Fails: not a function**



A function maps each allowed input to exactly one output:

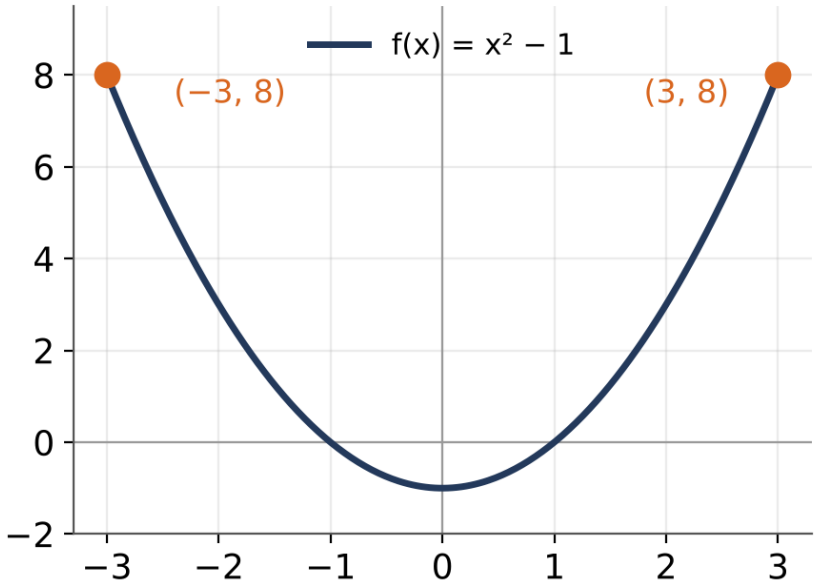
$$f : D \to \mathbb{R}, \quad x \mapsto f(x)$$

FUNCTION

Mathematica

```
Clear[x, f]
f[x_] := x^2 - 1
f[3]
f[a]
Plot[f[x], {x, -3, 3}]
```

For $f(x) = x^2 - 1$: $f(3) = 8$ and $f(-3) = 8$. Two inputs may share an output; one input never has two outputs.



The graph is the set of points {(x, f(x)) : x ∈ D}.

How Mathematica reads Mathematics

Minimal syntax you need today

```
Sin[x]     (* sin x        *)
x^2        (* x squared    *)
2*x        (* 2 times x    *)
Pi         (* π            *)
Exp[x]     (* e^x          *)
Sqrt[x]    (* square root  *)
```

Mathematics	Mathematica
sin x	`Sin[x]`
x^2	`x^2`
2x	`2*x  or  2 x`
π	`Pi`
e^x	`Exp[x]`
√x	`Sqrt[x]`

Two rules to remember

- Square brackets [ ] mean function input. Parentheses ( ) do not.
- Built-in names start with a capital letter: Sin, not sin.

Exact vs Approximate: Two kinds of answers

Try these

$\frac{1}{3} + \frac{1}{6}$

$\frac{1.0}{3} + \frac{1}{6}$

```
Sqrt[2]
N[Sqrt[2], 12]
Pi
N[Pi, 12]
```

Input	Output	Kind
$\frac{1}{3} + \frac{1}{6}$	½	Exact
$\frac{1.0}{3} + \frac{1}{6}$	0.5	Approximate
`Sqrt[2]`	√2	Exact
`N[Sqrt[2], 12]`	1.41421356237	Approximate

- Exact expressions keep the mathematical structure.
- Decimals are useful when you need a measurement or a numerical comparison.

Watch the decimal point

A single decimal in the input (1.0 instead of 1) tells Mathematica to work approximately for the whole calculation.

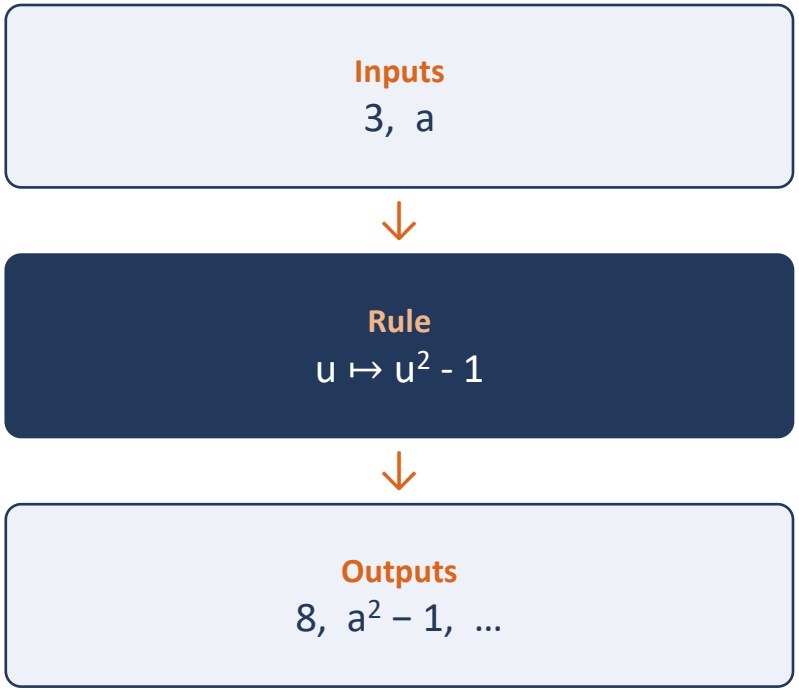
Defining a function: Why the underscore matters

Mathematica

```
Clear[x, f]
f[x_] := x^2 - 1
f[3]
f[a]
```

x_ means: "accept any expression here and call it x".

So, the same rule can receive a number, a symbol or an expression.



Plot: What is the command asking Mathematica to do?

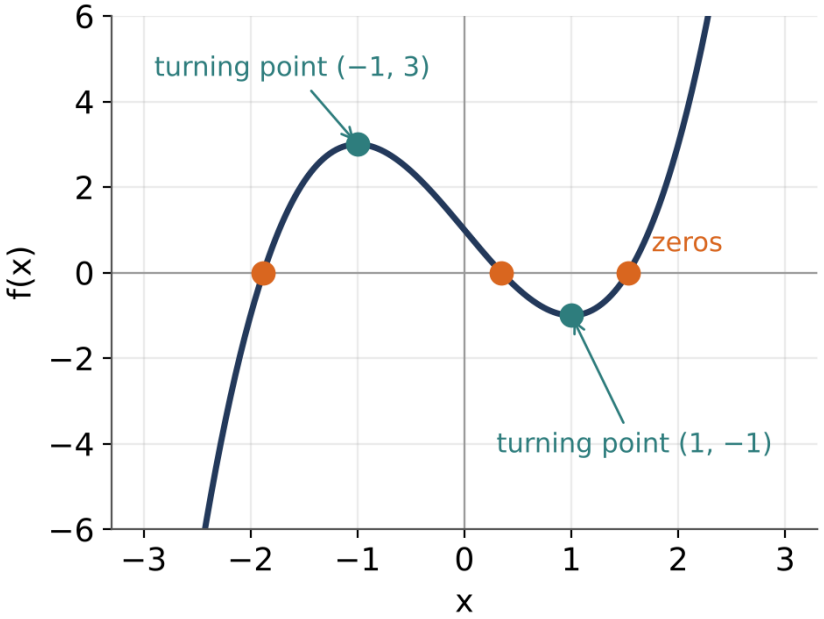
Simplest useful version

```
f[x_] := x^3 - 3*x + 1
Plot[f[x], {x, -3, 3}]
```

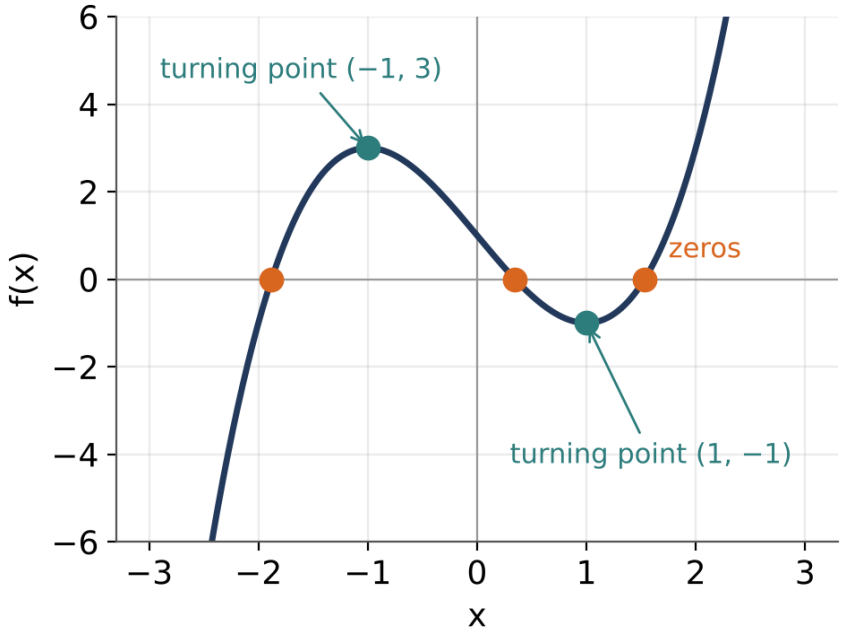
Then improve communication

```
Plot[f[x], {x, -3, 3},
  AxesLabel -> {"x", "f(x)"},
  GridLines -> Automatic]
```

Plot samples the function across the interval and draws the resulting curve. The interval [−3, 3] is part of the mathematical question: a different window can hide or reveal features.



Reading a graph: Four questions before any calculus



From the graph of $f(x) = x^3 - 3x + 1$, estimate:

1. Where is $f(x) = 0$?
2. Where is $f(x) > 0$ or $f(x) < 0$?
3. Where does f increase or decrease?
4. Where are the turning points?

Evidence, not proof

The graph gives evidence and intuition. Later, calculus gives exact tests.

Domain: Where is the formula actually allowed?

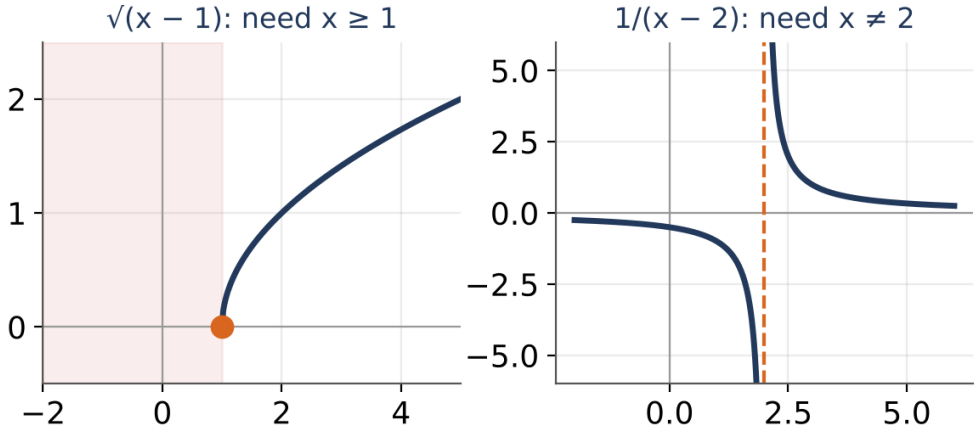
Mathematica

```
Plot[Sqrt[x - 1], {x, -2, 5}]
FunctionDomain[Sqrt[x - 1], x]
Plot[1/(x-2), {x, -2, 6},
  Exclusions -> {2}]
```

$\sqrt{(x-1)}$: $x - 1 \geq 0 \Rightarrow x \geq 1$

$\frac{1}{(x-2)}$: $x - 2 \neq 0 \Rightarrow x \neq 2$

Exclusions -> {2} stops Mathematica from joining the two branches with a false vertical line.



Problem 1: Can a function jump?

Simplest code

```
f[x_] := Piecewise[{
   {x^2, x < 1},
   {3,   x >= 1}
  }]
Plot[f[x], {x, -1, 3}]
```

The function is

$$f(x) = x^2 \text{ for } x < 1, \quad f(x) = 3 \text{ for } x \geq 1.$$

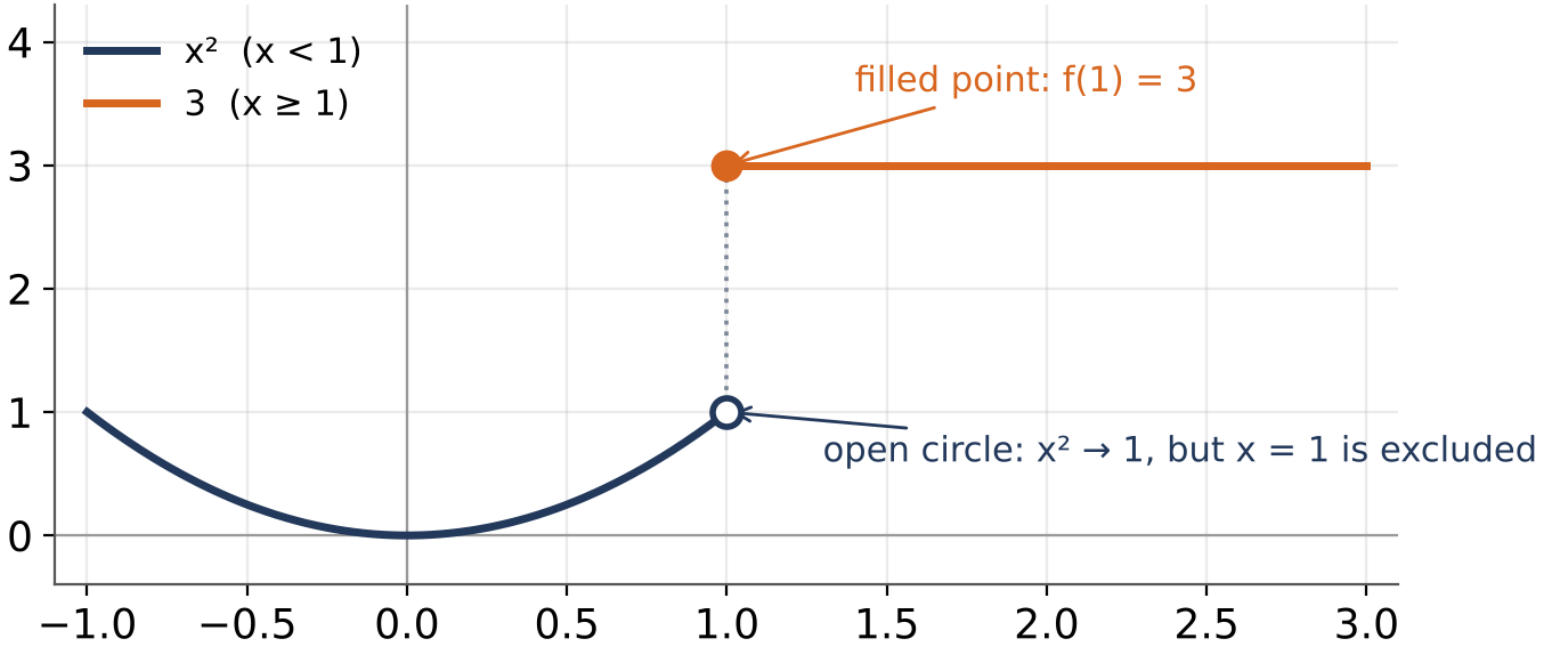
At x = 1 there are three different questions:

Question	Value
lim f(x) as x → 1^- (LHL)	1
lim f(x) as x → 1^+ (RHL)	3
f(1)	3

Conclusion

The one-sided limits disagree, so $\lim_{x \to 1} f(x)$ does not exist. Therefore, f is not continuous at x = 1.

Visualizing the jump: What the picture means



Open circle: The left formula approaches to 1 but does not include x = 1.

Filled point: The actual function value is f(1) = 3.

Here, the dotted gap is the jump.

Verify the picture symbolically

Mathematica

```
Limit[f[x], x -> 1,
  Direction -> "FromBelow"]
Limit[f[x], x -> 1,
  Direction -> "FromAbove"]
f[1]
```

Continuity at x = a requires all three to agree:

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a)$$

Here the left limit is 1 and the right limit is 3. Since $1 \neq 3$, the discontinuity is a **jump discontinuity**.

Workflow

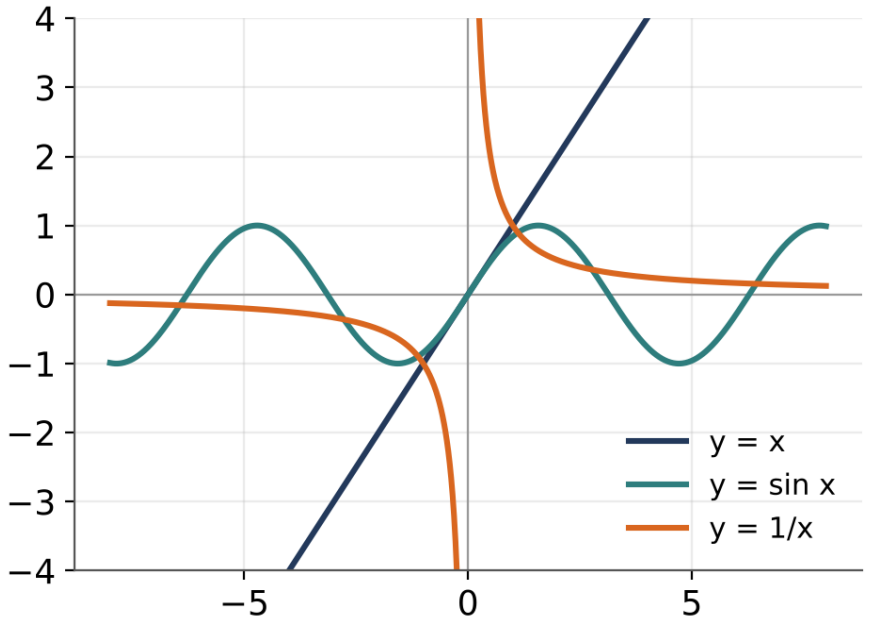
Plot → **Predict** → **Compute Limits** → **Decide Continuity**

Problem 2: Compare three very different functions

One plot, three expressions

```
Plot[{x, Sin[x], 1/x},
  {x, -8, 8},
  PlotRange -> {-4, 4},
  PlotLegends ->
    {"x", "sin x", "1/x"}]
```

Function	Domain	Large |x| behavior
x	$\mathbb{R}$	Unbounded
sin x	$\mathbb{R}$	Oscillates in [−1, 1]
$\frac{1}{x}$	$\mathbb{R} \setminus \{0\}$	Tends to 0



Same coordinate plane, very different behavior. The graph should help us to compare, not just decorate.

Why $\frac{1}{x}$ breaks at the origin

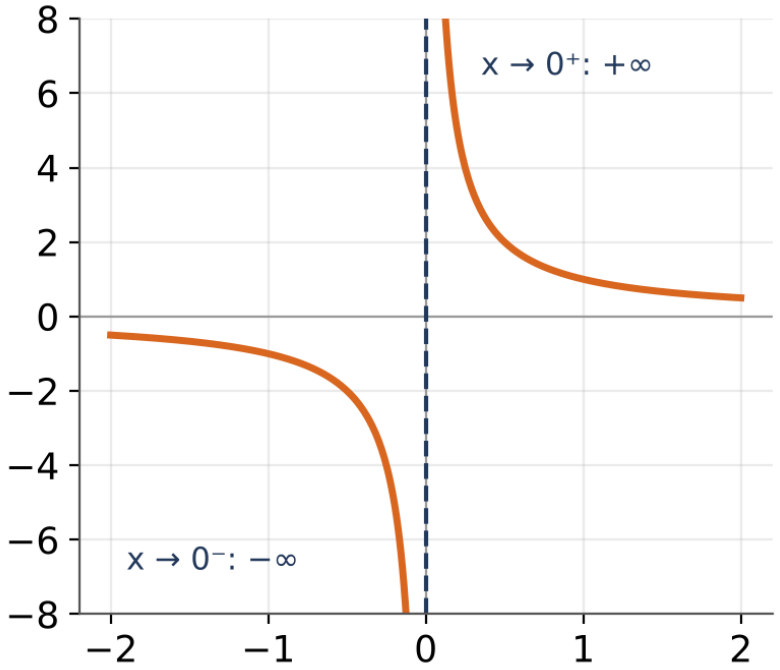
Zoom near zero

```
Plot[1/x, {x, -2, 2},
  PlotRange -> {-8, 8},
  Exclusions -> {0},
  GridLines -> Automatic]
```

$h(x) = \frac{1}{x}$ requires division by x. At x = 0 division is undefined. Moreover,

$$\lim_{x \to 0^-} \frac{1}{x} = -\infty, \quad \lim_{x \to 0^+} \frac{1}{x} = +\infty.$$

The line x = 0 is a **vertical asymptote**.

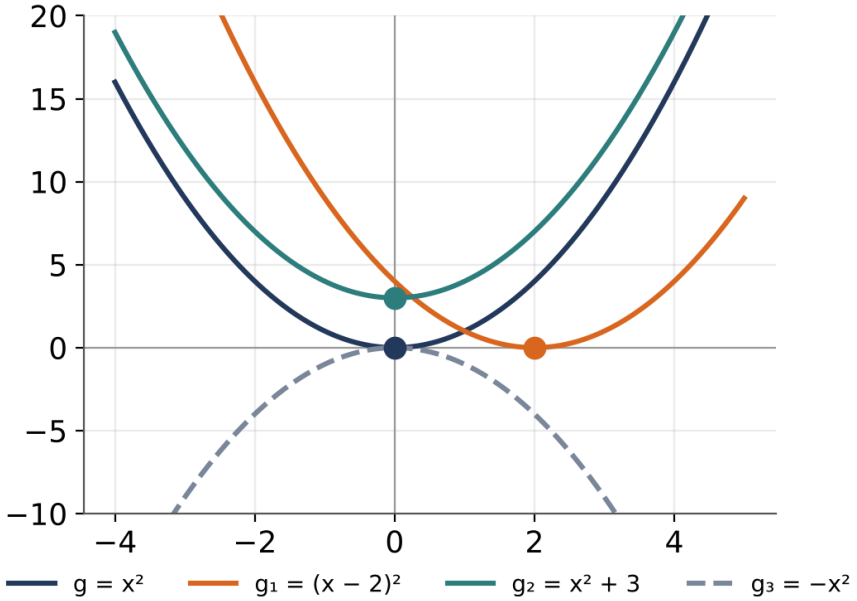


Problem 3: Transformations of g(x) = x^2

Compare on one graph

```
Plot[{x^2, (x-2)^2,
      x^2 + 3, -x^2},
  {x, -4, 5},
  PlotRange -> {-10, 20},
  PlotLegends ->
    {"g", "g1", "g2", "g3"}]
```

Function	Change	Vertex
$g_1 = (x - 2)^2$	shift right 2	(2, 0)
$g_2 = x^2 + 3$	shift up 3	(0, 3)
$g_3 = -x^2$	reflect in x-axis	(0, 0)



Why does $(x - 2)^2$ move to the right?

The vertex of $g(x) = x^2$ occurs where the squared quantity is zero: $x = 0$.

For $g_1(x) = (x - 2)^2$, the square is zero when

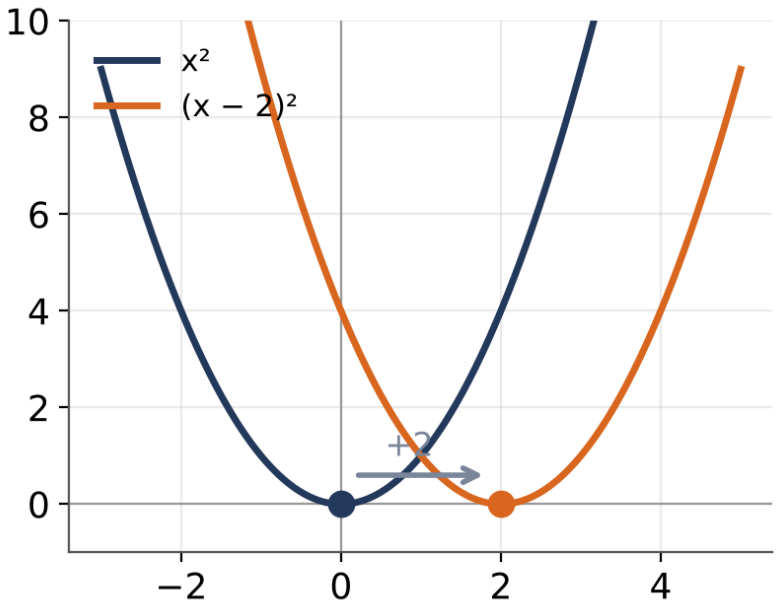
$$x - 2 = 0 \quad \Rightarrow \quad x = 2.$$

So, the vertex moves from (0, 0) to (2, 0).

Common Confusion

A minus sign inside the bracket moves the graph right, not left.

Ask: what x makes the inside zero?



Problem 4: Exploring a Rational Function

Simplest code

```
h[x_] := (x + 1)/(x - 2)
Plot[h[x], {x, -6, 8},
  Exclusions -> {2},
  PlotRange -> {-6, 6}]
```

Useful symbolic checks

```
Solve[h[x] == 0, x]
h[0]
Limit[h[x], x -> Infinity]
```

$$h(x) = \frac{(x + 1)}{(x - 2)}$$

Domain: $x \neq 2$. Vertical asymptote: $x = 2$.

Rewrite:

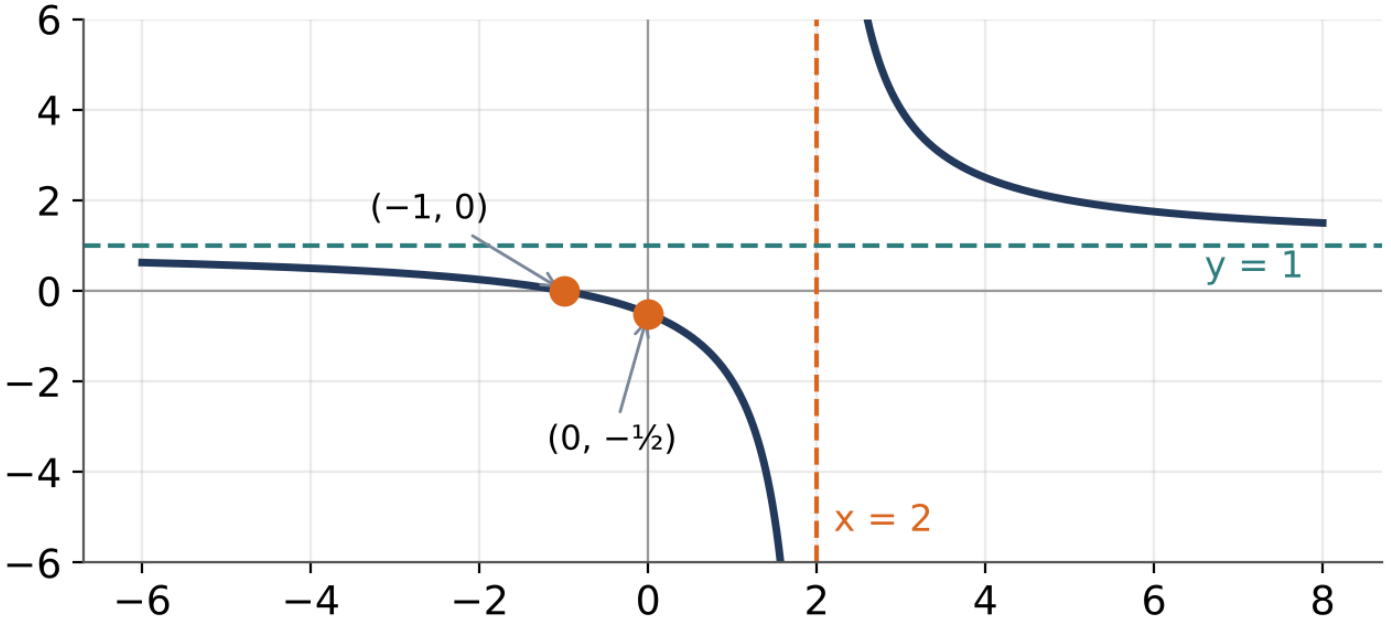
$$\frac{(x + 1)}{(x - 2)} = 1 + \frac{3}{(x-2)}$$

So, $h(x) \to 1$ as $|x| \to \infty$:

horizontal asymptote **y = 1**.

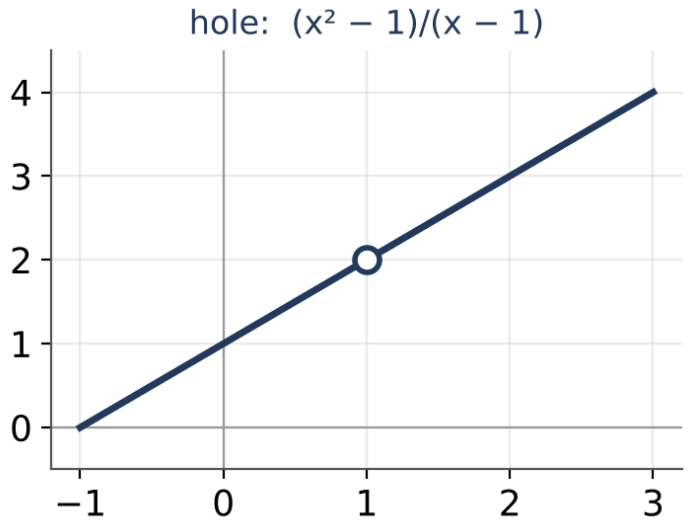
Intercepts: $x = -1$ and $h(0) = -1/2$.

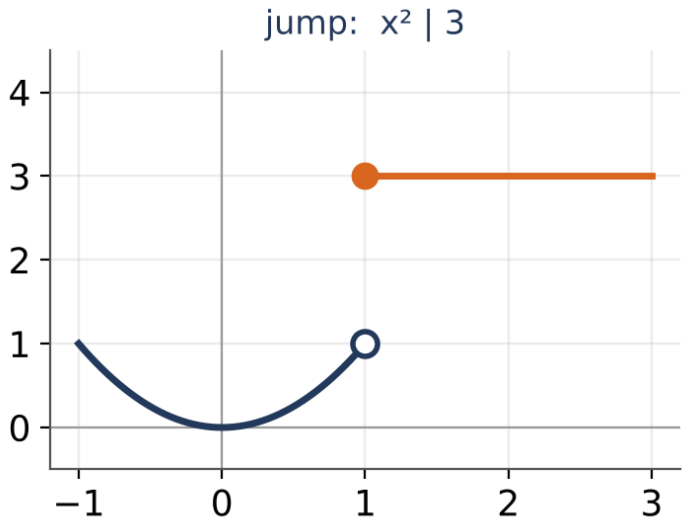
The rational graph explained, not just plotted



Near x = 2 the denominator becomes very small, so |h(x)| becomes very large. Far from x = 2 the correction $\frac{3}{(x-2)}$ becomes small, so the graph approaches y = 1.

Removable Discontinuity vs Jump Discontinuity





A hole

$r(x) = \frac{(x^2-1)}{(x-1)} = x + 1$ for $x \neq 1$. The limit at 1 is 2, but the original expression is undefined there.

Same limiting height; the point is missing.

A jump

The left limit is 1 and the right limit is 3. The two sides approach different heights, so no single value can fix it.

Lists and Tables: Connect discrete values to a graph

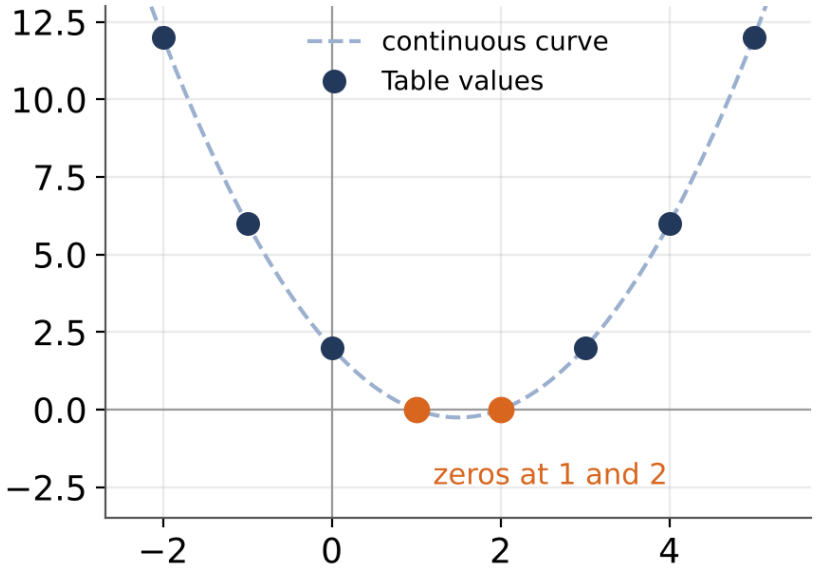
Mathematica

```
values = Table[
   {x, x^2 - 3*x + 2},
   {x, -2, 5}]
TableForm[values]
ListPlot[values,
  PlotMarkers -> Automatic]
```

A graph contains infinitely many inputs; a table samples selected ones.

$y = x^2 - 3x + 2 = (x - 1)(x - 2)$

Habit: Predict *one feature before running the code, then check the output agrees.*



Guided activity: Investigate one function

Start with only this

```
Clear[x, p]
p[x_] := x^4 - 4*x^2 + x + 1
Plot[p[x], {x, -3, 3}]
```

Tip

Compare your graph estimates with the NSolve output. How close were you?

PROBLEM-01:

For $p(x) = x^4 - 4x^2 + x + 1$, work in this order:

1. predict the end behavior;
2. plot on [−3, 3];
3. make an integer-value table;
4. estimate the x-intercepts;
5. use NSolve[p[x] == 0, x] to refine the roots.

Common Mistakes: Diagnose the Mathematical Consequence

Mistake	Example	What goes wrong
Wrong brackets	`Sin(x)`	Not read as the built-in sine function.
Missing pattern	`f[x] := x^2`	Does not define a general one-argument rule.
Old value stored	`x = 2 before plotting`	x is replaced before Plot can use it as a variable.
Assignment vs equation	`x = 3 in Solve`	= stores a value; == states an equation.
Ignoring domain	`plotting 1/(x - 2) blindly`	A graph may be misread near asymptotes.

Debug the syntax, but also ask what the syntax means mathematically.

LAB REPORT-01

A. Continuity

f(x) = x + 2 for x < 0, x^2 for x ≥ 0. Is f continuous at x = 0?

B. Transformations

Compare |x|, |x − 4| and |x| + 1. Describe each change.

C. Rational function

f(x) = $\frac{(x^2 - 3)}{(x - 1)}$. Find the domain and intercepts, then plot.

D. Create

Sketch a function continuous everywhere except x = 2. Is it removable, jump or infinite?

E. Explain

In your own words: the difference between a removable and a jump discontinuity.

- *Plot first*
- *Compute second*
- *Interpret last*

A repeatable workflow for every Mathematica lab

1. Understand the Question
2. Write the Mathematics
3. Translate to Code
4. Visualize / Compute
5. Check Reasonableness
6. Interpret in Words

Mathematica should shorten computation, not shorten thinking.

Task

Code

```
Clear[x, h]
h[x_] := (x + 1)/(x - 2)
Plot[h[x], {x, -6, 8},
  Exclusions -> {2}]
```

PROBLEM-02:

1. What is the domain of h?
2. What are its vertical and horizontal asymptotes?
3. What are its intercepts?
4. What does the graph do near x = 2?
5. Which Mathematica command would you use to verify the horizontal asymptote?